

# Day 102 — TNPSC Group 2A Study Material

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## Part A – General Studies: Current Affairs in General Science (9): India's nOPV2 Polio Vaccine Gains WHO Prequalification

### READING MATERIAL

*Covers a 2026 global-health milestone: an Indian manufacturer's novel oral polio vaccine achieving WHO prequalification.*

**Q:** Which Indian company's novel oral polio vaccine type 2 (nOPV2) received WHO prequalification, and when?

**A:** Biological E Limited (BioE) — prequalified by WHO on 12 February 2026.

**Q:** What technology partnership enabled Biological E's full-scale manufacturing of nOPV2?

**A:** A technology transfer from PT Bio Farma (Persero), Indonesia — expanding BioE's role from fill-finish activities (using PT Bio Farma's bulk vaccine) to full in-house bulk manufacturing.

**Q:** How many doses of nOPV2 has Biological E already contributed to the global stockpile, and what is its projected additional annual capacity after this prequalification?

**A:** 700 million doses contributed since March 2024; an additional 600 million doses per year projected with the new prequalification.

**Q:** What is nOPV2's significance for global health, and which 2 manufacturers now produce it at full scale?

**A:** It is a critical tool for stopping outbreaks of the type-2 variant poliovirus. With this prequalification, Biological E (India) becomes the SECOND full manufacturer of nOPV2, joining PT Bio Farma (Indonesia) — together providing diversified, more stable global production capacity.

### MOCK TEST (WITH ANSWERS)

1. Which Indian company's nOPV2 (novel oral polio vaccine type 2) received WHO prequalification on 12 February 2026?

- A. Bharat Biotech
- B. Serum Institute of India
- C. Biological E Limited ✓**
- D. Panacea Biotec

*Explanation:* Biological E Limited. [medium, web-research]

2. Biological E's nOPV2 manufacturing capability came via a technology transfer from which country's manufacturer?

- A. Indonesia (PT Bio Farma) ✓**
- B. United States (Pfizer)
- C. United Kingdom (GSK)
- D. Japan (Takeda)

*Explanation:* Indonesia (PT Bio Farma). [hard, web-research]

3. Since March 2024, Biological E has contributed how many doses of nOPV2 to the global stockpile?

- A. 300 million
- B. 500 million
- C. 700 million ✓**
- D. 900 million

*Explanation:* 700 million doses. [hard, web-research]

4. nOPV2 is specifically a tool for stopping outbreaks of which poliovirus variant?

- A. Type 1
- B. Type 2 ✓**
- C. Type 3
- D. Wild poliovirus only

*Explanation:* Type 2 variant poliovirus. [medium, web-research]

5. With this WHO prequalification, Biological E becomes the world's \_\_\_\_ full manufacturer of nOPV2.

- A. 1st

**B. 2nd ✓**

C. 3rd

D. 4th

*Explanation:* 2nd (after PT Bio Farma, Indonesia). [medium, web-research]

## Part B – Aptitude & Mental Ability: Simple Interest

### READING MATERIAL

*Simple Interest is 1 of 10 named Aptitude sub-skills. Real exam evidence shows TNPSC favors word-problem variants (doubling time, comparing two loans, fractional rates with month-based time) over the bare 'calculate SI' formula alone.*

**Q:** Formula: What is the Simple Interest formula, and how do you get the doubling time from it?

**A:**  $SI = (\text{Principal} \times \text{Rate} \times \text{Time}) / 100$ . A sum **DOUBLES** when SI equals the Principal itself:  $P = (P \times R \times T) / 100 \rightarrow T = 100/R$  years. So at 8% per annum, doubling time =  $100/8 = 12.5$  years = 150 months.

**Q:** Real exam (2024): In how many months will Rs.2,000 double itself at 8% per annum simple interest?

**A:**  $T = 100/R = 100/8 = 12.5$  years =  $12.5 \times 12 = 150$  months.

**Q:** Real exam (2022): What is the simple interest on Rs.1,000 for one year at 10% per annum?

**A:**  $SI = (1000 \times 10 \times 1) / 100 = \text{Rs.}100$ .

**Q:** Real exam (2024): Find the simple interest on Rs.68,000 at  $16\frac{2}{3}\%$  per annum for 9 months.

**A:** Convert the mixed-fraction rate:  $16\frac{2}{3}\% = 50/3\%$ . Convert months to years: 9 months =  $9/12 = 3/4$  year.  $SI = 68,000 \times (50/3) \times (3/4) / 100 = 68,000 \times 12.5/100 = \text{Rs.}8,500$ . (The 3s cancel neatly between the rate's denominator and the time's numerator — recognizing that shortcut avoids messy fraction arithmetic.)

**Q:** Real exam (2025): A borrows Rs.16,000 at 12% p.a. simple interest, B borrows Rs.12,000 at 18% p.a. simple interest. In how many years will their total amounts owed (principal+interest) become equal?

**A:** Set up the equality:  $16000 + 16000(0.12)t = 12000 + 12000(0.18)t \rightarrow 16000 + 1920t = 12000 + 2160t \rightarrow 4000 = 240t \rightarrow t = 16\frac{2}{3}$  years.

**Q:** Method — finding the rate when given principal, time, and final amount: A sum of Rs.5,000 becomes Rs.6,200 in 3 years at simple interest. Find the rate.

**A:**  $SI = \text{Amount} - \text{Principal} = 6200 - 5000 = 1200$ . Rearranged formula:  $R = (SI \times 100) / (P \times T) = (1200 \times 100) / (5000 \times 3) = 120000/15000 = 8\%$  per annum.

### MOCK TEST (WITH ANSWERS)

1. In how many years will a sum double itself at 5% per annum simple interest?

- A. 10 years
- B. 15 years
- C. 20 years ✓**
- D. 25 years

*Explanation:*  $T = 100/R = 100/5 = 20$  years. [easy, authored]

2. Find the simple interest on Rs.2,500 for 1 year at 12% per annum.

- A. Rs.250
- B. Rs.275
- C. Rs.300 ✓**
- D. Rs.320

*Explanation:*  $SI = (2500 \times 12 \times 1) / 100 = \text{Rs.}300$ . [easy, authored]

3. Find the simple interest on Rs.54,000 at  $13\frac{1}{3}\%$  per annum for 6 months.

- A. Rs.3,600 ✓**
- B. Rs.3,000
- C. Rs.3,300
- D. Rs.4,000

*Explanation:*  $13\frac{1}{3}\% = 40/3\%$ . 6 months =  $1/2$  year.  $SI = 54000 \times (40/3) \times (1/2) / 100 = 54000 \times 40/600 = 54000 \times (1/15) = 3,600$ . [hard, authored]

4. C borrows Rs.20,000 at 10% p.a. simple interest, D borrows Rs.15,000 at 15% p.a. simple interest. In how many years will their total amounts owed become equal?

- A. 8 years
- B. 10 years

**C. 20 years ✓**

D. 25 years

*Explanation:*  $20000 + 2000t = 15000 + 2250t \rightarrow 5000 = 250t \rightarrow t = 20$  years. [hard, authored]

5. A sum of Rs.8,000 becomes Rs.9,440 in 3 years at simple interest. Find the rate per annum.

A. 5%

**B. 6% ✓**

C. 6.5%

D. 7%

*Explanation:*  $SI = 9440 - 8000 = 1440$ .  $R = (1440 \times 100) / (8000 \times 3) = 144000 / 24000 = 6\%$ . [medium, authored]

6. What sum will yield Rs.900 as simple interest in 3 years at 6% per annum?

A. Rs.4,000

**B. Rs.5,000 ✓**

C. Rs.5,500

D. Rs.6,000

*Explanation:*  $P = (SI \times 100) / (R \times T) = (900 \times 100) / (6 \times 3) = 90000 / 18 = \text{Rs.}5,000$ . [medium, authored]

## Part C – General English: Poem: "Never Trust a Mirror" — Erin Hanson

### READING MATERIAL

*A poem arguing that mirrors only show surface appearance, not true self-worth. Real coaching-site evidence tests theme, symbolism, and a couple of figures of speech.*

**Q:** Who wrote "Never Trust a Mirror", and what is its central theme?

**A:** Erin Hanson. Central theme: self-worth comes from within, not from outer appearance.

**Q:** What does the mirror symbolize in the poem?

**A:** Outer appearance and illusion — the mirror shows only surface reflection, not a person's true worth or character.

**Q:** What does "skin deep" mean, as used in the poem?

**A:** Superficial, or only outward — describing something that has no depth beyond physical appearance.

**Q:** Identify the figure of speech: "The mirror always lies."

**A:** Personification — a mirror (an inanimate object) is given the human capacity to 'lie'.

**Q:** Identify the figure of speech: "When your eyelids flutter."

**A:** Imagery — a vivid, visual sensory description.

### MOCK TEST (WITH ANSWERS)

1. "Never Trust a Mirror" was written by:

- A. Maya Angelou
- B. Robert Frost
- C. Erin Hanson ✓**
- D. W.H. Davies

*Explanation:* Erin Hanson. [easy, web-research]

2. The main theme of the poem is:

- A. Physical beauty
- B. The truth in reflections
- C. Self-worth comes from within ✓**
- D. Dreams and goals

*Explanation:* Self-worth comes from within. [easy, web-research]

3. What does the mirror symbolize in the poem?

- A. Confidence
- B. Truth
- C. Outer appearance and illusion ✓**
- D. Technology

*Explanation:* Outer appearance and illusion. [medium, web-research]

4. "The mirror always lies" uses which figure of speech?

- A. Simile
- B. Hyperbole
- C. Personification ✓**
- D. Irony

*Explanation:* Personification — the mirror given the human ability to 'lie'. [medium, web-research]

5. In the poem, "skin deep" means:

- A. Very deep
- B. Superficial or only outward ✓**
- C. Emotionally strong
- D. Spiritually rich

*Explanation:* Superficial/only outward. [medium, web-research]

6. "When your eyelids flutter" is an example of which poetic device?

A. Metaphor

B. Alliteration

**C. Imagery ✓**

D. Irony

*Explanation:* Imagery — vivid visual description. [medium, web-research]